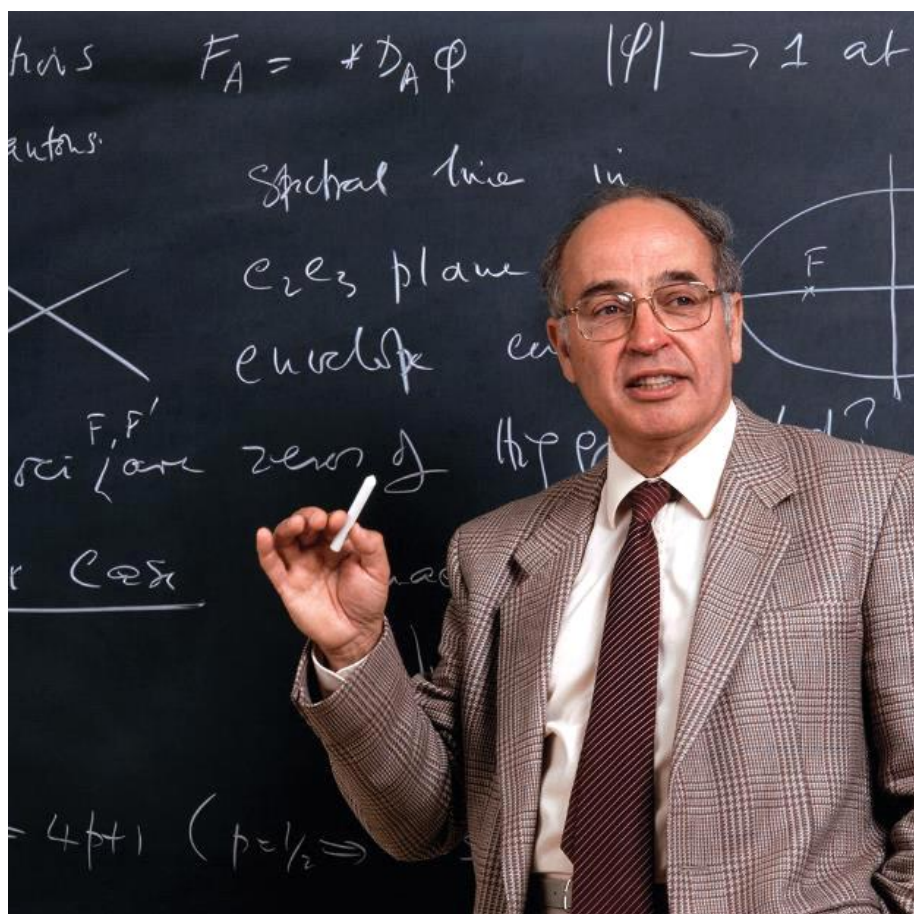
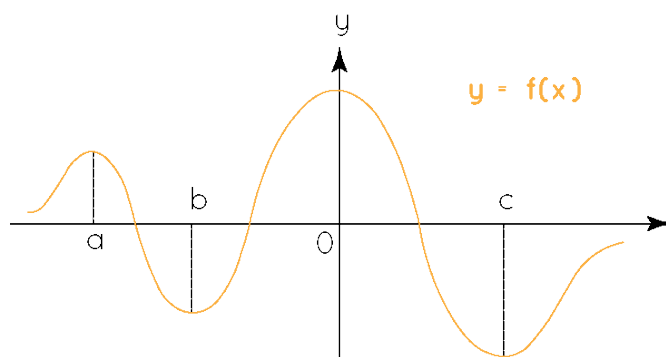


Lec-03-Week-01

Monday, 17 August 2026 10:51 am

Maxima and Minima

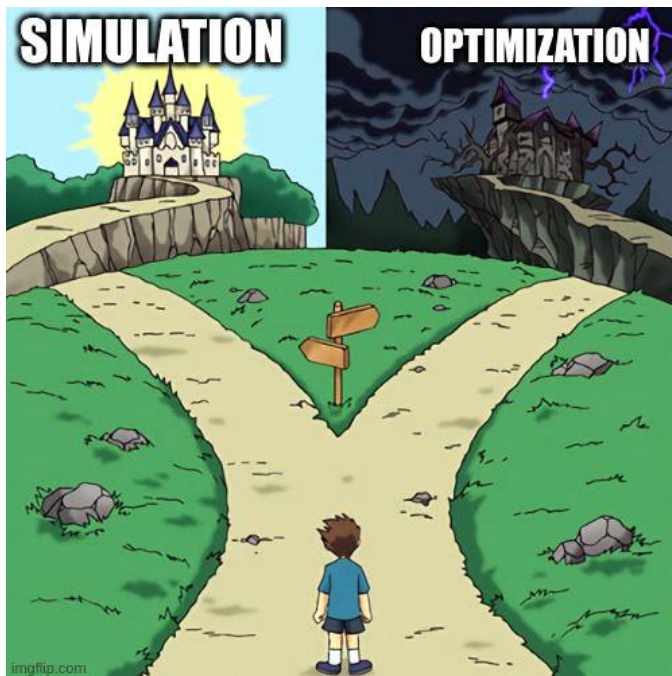


ALGEBRA IS THE OFFER MADE BY THE
DEVIL TO THE MATHEMATICIAN. THE
DEVIL SAYS: I WILL GIVE YOU THIS
POWERFUL MACHINE, IT WILL ANSWER
ANY QUESTION YOU LIKE. ALL YOU
NEED TO DO IS GIVE ME YOUR SOUL:
GIVE UP GEOMETRY AND YOU WILL
HAVE THIS MARVELLOUS MACHINE.

- MICHAEL ATIYAH -

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CRITICAL POINTS: LOCAL EXTREMA AND SADDLE POINTS

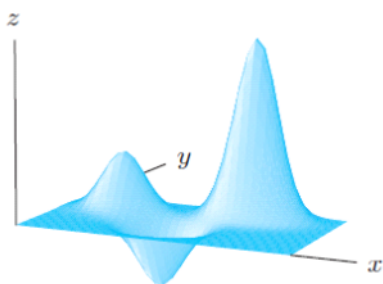


Figure 15.1: Local and global extrema for a function of two variables on $0 \leq x \leq a$, $0 \leq y \leq b$

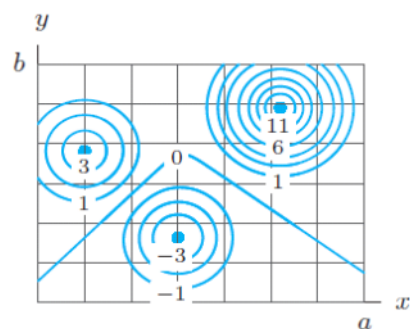
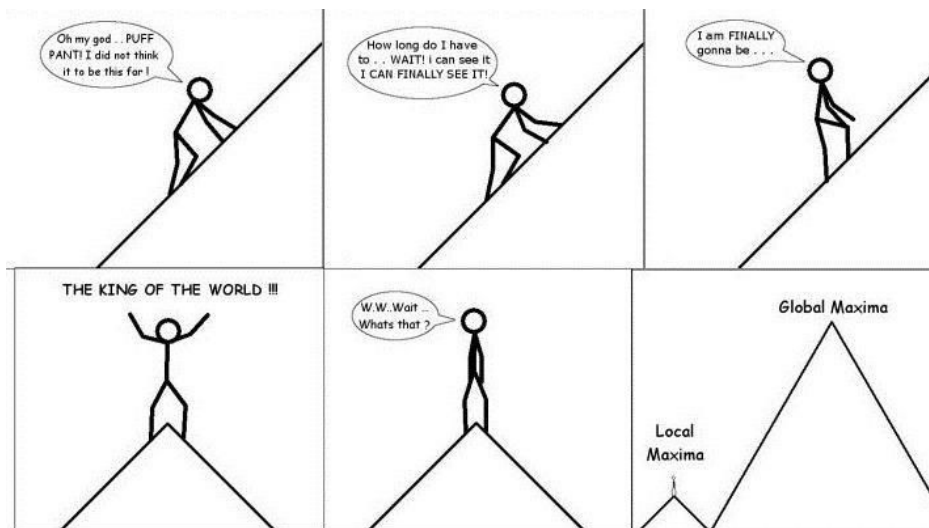


Figure 15.2: Contour map of the function in Figure 15.1

- f has a **local maximum** at the point P_0 if $f(P_0) \geq f(P)$ for all points P near P_0 .
- f has a **local minimum** at the point P_0 if $f(P_0) \leq f(P)$ for all points P near P_0 .

nobody:

algorithm: ah yes the solution



Just because there is nothing left to derive in life $\frac{dl}{dt} = 0$ doesn't mean you are the best. Know the difference.
Don't be fooled

Points where the gradient is either $\vec{0}$ or undefined are called **critical points** of the function.

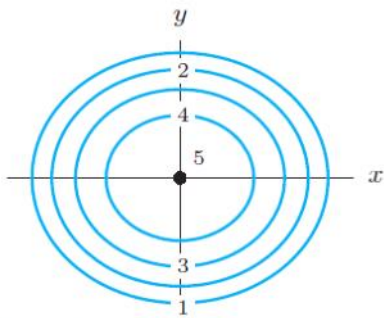


Figure 15.3: Contour diagram around a local maximum of a function

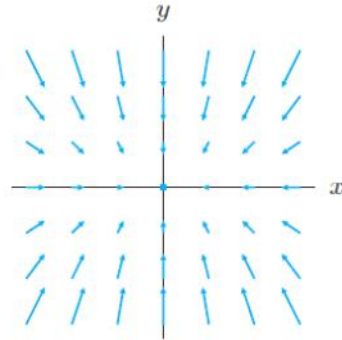


Figure 15.4: Gradients pointing toward the local maximum of the function in Figure 15.3

Example 1 Find and analyze the critical points of $f(x, y) = x^2 - 2x + y^2 - 4y + 5$.

Solution To find the critical points, we set both partial derivatives equal to zero:

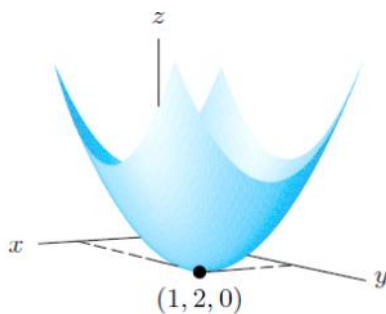
$$f_x(x, y) = 2x - 2 = 0$$

$$f_y(x, y) = 2y - 4 = 0.$$

Solving these equations gives $x = 1$, $y = 2$. Hence, f has only one critical point, namely $(1, 2)$. To see the behavior of f near $(1, 2)$, look at the values of the function in Table 15.1.

$$f(x, y) = x^2 - 2x + y^2 - 4y + 5 = (x - 1)^2 + (y - 2)^2.$$

		x				
		0.8	0.9	1.0	1.1	1.2
y	1.8	0.08	0.05	0.04	0.05	0.08
	1.9	0.05	0.02	0.01	0.02	0.05
	2.0	0.04	0.01	0.00	0.01	0.04
	2.1	0.05	0.02	0.01	0.02	0.05
	2.2	0.08	0.05	0.04	0.05	0.08



Example 2 Find and analyze any critical points of $f(x, y) = -\sqrt{x^2 + y^2}$.

Solution We look for points where $\text{grad } f = \vec{0}$ or is undefined. The partial derivatives are given by

$$f_x(x, y) = -\frac{x}{\sqrt{x^2 + y^2}},$$
$$f_y(x, y) = -\frac{y}{\sqrt{x^2 + y^2}}.$$

These partial derivatives are never simultaneously zero, but they are undefined at $x = 0, y = 0$. Thus, $(0, 0)$ is a critical point and a possible extreme point. The graph of f (see Figure 15.6) is a cone, with vertex at $(0, 0)$. So f has a local and global maximum at $(0, 0)$.

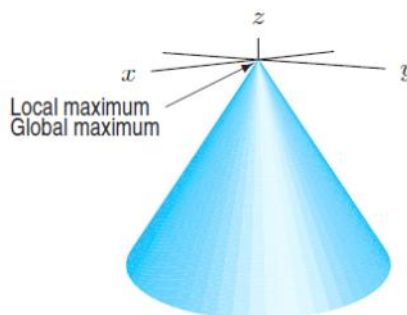


Figure 15.6: Graph of $f(x, y) = -\sqrt{x^2 + y^2}$

Example 3 Find and analyze any critical points of $g(x, y) = x^2 - y^2$.

Solution To find the critical points, we look for points where both partial derivatives are zero:

$$g_x(x, y) = 2x = 0$$
$$g_y(x, y) = -2y = 0.$$

Solving gives $x = 0, y = 0$, so the origin is the only critical point.

Figure 15.7 shows that near the origin g takes on both positive and negative values. Since $g(0, 0) = 0$, the origin is a critical point which is neither a local maximum nor a local minimum. The graph of g looks like a saddle.

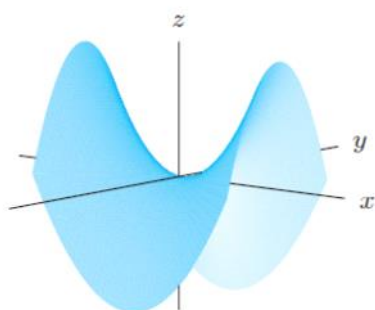


Figure 15.7: Graph of $g(x, y) = x^2 - y^2$, showing saddle shape at the origin

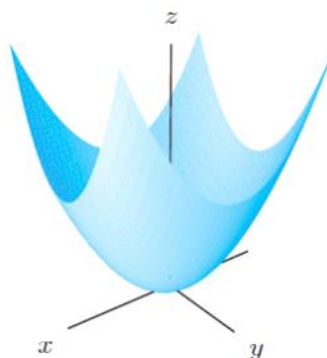


Figure 15.8: Graph of $h(x, y) = x^2 + y^2$, showing minimum at the origin



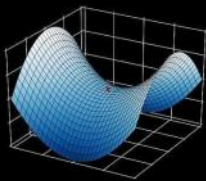
What others see

Potato chip



What i see

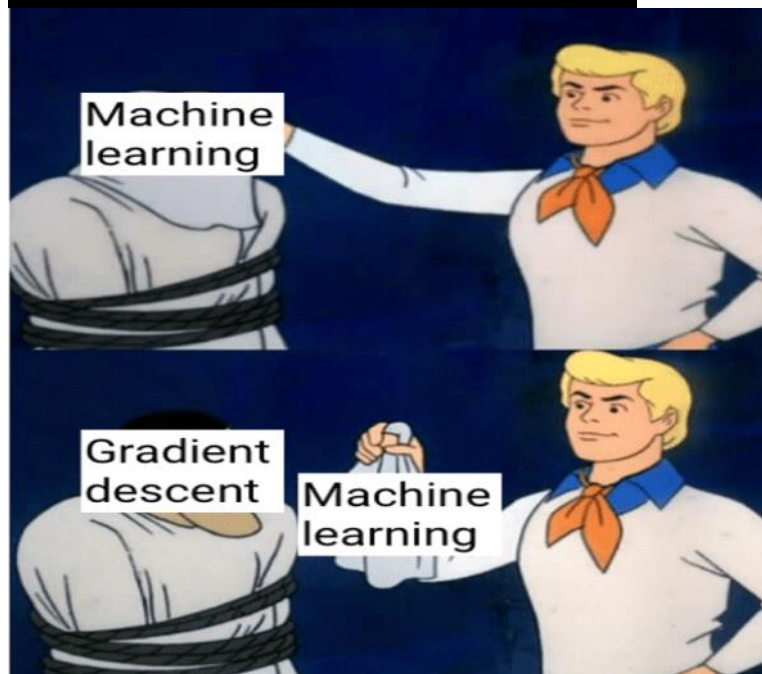
Gradient descent



Gradient descent is an iterative optimization algorithm for finding the minimum of a function.

$$x_{i+1} = x_i - \alpha \nabla f(x_i)$$

next position x_{i+1} opposite direction $-\nabla f(x_i)$ gradient at current position $\nabla f(x_i)$
 current position x_i step size (learning rate) α





Example 4 Find the local extrema of the function $f(x, y) = 8y^3 + 12x^2 - 24xy$.

Solution We begin by looking for critical points:

$$f_x(x, y) = 24x - 24y,$$

$$f_y(x, y) = 24y^2 - 24x.$$

Setting these expressions equal to zero gives the system of equations

$$x = y, \quad x = y^2,$$

which has two solutions, $(0, 0)$ and $(1, 1)$. Are these local maxima, local minima or neither? Figure 15.11 shows contours of f near the points. Notice that $f(1, 1) = -4$ and that there is no other -4 contour. The contours near $(1, 1)$ suggest that f has a local minimum at the point $(1, 1)$.

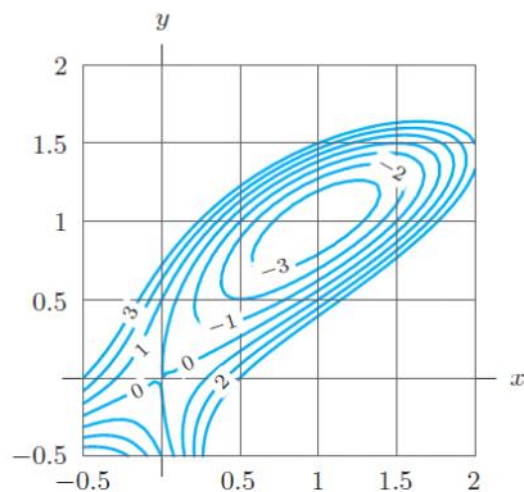


Figure 15.11: Contour diagram of $f(x, y) = 8y^3 + 12x^2 - 24xy$ showing critical points at $(0, 0)$ and $(1, 1)$

Second Derivative Test

Local Max

$$f'(c) = 0 \text{ and } f''(c) < 0$$



Local Min

$$f'(c) = 0 \text{ and } f''(c) > 0$$



Second-Derivative Test for Functions of Two Variables

Suppose (x_0, y_0) is a point where $\text{grad } f(x_0, y_0) = \vec{0}$. Let

$$D = f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - (f_{xy}(x_0, y_0))^2.$$

- If $D > 0$ and $f_{xx}(x_0, y_0) > 0$, then f has a local minimum at (x_0, y_0) .
- If $D > 0$ and $f_{xx}(x_0, y_0) < 0$, then f has a local maximum at (x_0, y_0) .
- If $D < 0$, then f has a saddle point at (x_0, y_0) .
- If $D = 0$, anything can happen: f can have a local maximum, or a local minimum, or a saddle point, or none of these, at (x_0, y_0) .

Example 6 Find the local maxima, minima, and saddle points of $f(x, y) = \frac{1}{2}x^2 + 3y^3 + 9y^2 - 3xy + 9y - 9x$.

Solution Setting the partial derivatives of f to zero gives

$$f_x(x, y) = x - 3y - 9 = 0,$$

$$f_y(x, y) = 9y^2 + 18y + 9 - 3x = 0.$$

Eliminating x gives $9y^2 + 9y - 18 = 0$, with solutions $y = -2$ and $y = 1$. The corresponding values of x are $x = 3$ and $x = 12$, so the critical points of f are $(3, -2)$ and $(12, 1)$. The discriminant is

$$D(x, y) = f_{xx}f_{yy} - f_{xy}^2 = (1)(18y + 18) - (-3)^2 = 18y + 9.$$

Since $D(3, -2) = -36 + 9 < 0$, we know that $(3, -2)$ is a saddle point of f . Since $D(12, 1) = 18 + 9 > 0$ and $f_{xx}(12, 1) = 1 > 0$, we know that $(12, 1)$ is a local minimum of f .



Figure 15.13: Local minimum:
 $D > 0$ and $a > 0$



Figure 15.14: Local maximum:
 $D > 0$ and $a < 0$



Figure 15.15: Saddle point:
 $D < 0$



Figure 15.16: Parabolic
cylinder: $D = 0$